

Family 1.tf: the terrestrial and coastal fine-granularity observation family

a dataset design note — stations, gauges, radars, gliders, fires,
daily 5 km land fields, and the tile catalogue
raw observations first; the licence decides what may be published; the tile tier is a
catalogue until its codec exists

blauewelt/earth — design note

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Abstract

Family 1.tf is the land-and-coast sibling of family 1.gf: observation stores at **ten kilometres or finer and five days or finer** over the continents and a coastal band, built on family 10's registry, tiers and token. Land and coast differ from the open ocean in three ways that shape the design. Most of the fine record is *points* — 38,870 hourly weather stations, 132,437 daily ones, 2,932 radiosonde sites, 6,474 tide-gauge records, 1,345 buoys, hourly coastal current maps from high-frequency radar, 3,660 glider deployments, 700 flux towers, and half a billion satellite fire detections — and they fit tier P with one widening of the time column, at about 0.65 TB for everything that may be published. The daily *fields* at 0.05° (MODIS and VIIRS land-surface temperature, reflectance and snow; IMS 1 km snow; CHIRPS 0.05° pentad rainfall) are 20–30 % valid over land and need the sharded, tiled, compressed gridded layout family 1.gf introduces. And the finest imagery — Sentinel-1 and -2, Landsat and HLS, NISAR, 300 m OLCI, 375 m VIIRS — is 0.05–3 petabytes a year per sensor and can never be pixels: it enters as a *catalogue* store now (one row per scene-date, its footprint, time, cloud and validity) and as cached local-codec tokens later, per E-078's tier T, with the first token build coastal (E-078c). The research also drew a licence line that the ocean family did not need: the soil-moisture network (ISMN), the river-discharge archive (GRDC, except its open Caravan subset), the European station set (ECA&D) and HadISD forbid redistribution and are therefore not in this family however good they are. Every candidate was checked against its producer on 2026-09-16; three sources changed under the design in the last year (ISD frozen and replaced by GHCNv; Suomi-NPP ending 2026-11-01; Sentinel-1A retired), and the family is built from the successors. The note gives the ledger, the coastal-band definition, the per-store arithmetic, a three-phase ladder (phase A $\approx$ 0.6 TB, eleven stores counting the eight scene catalogues as one, fetched without an account, with an Earthdata login on a hosted runner, or with a Copernicus account for the Sentinel catalogues), what each store is *for* against the four goals — mostly forcing, coastal sea level, and the read-out side of “predict everything” — and the decisions left to Chris.

1 Scope: the rule, the two domains, and the coastal band

The rule is family 1.gf's: an observation with spatial support ≤ 10 km (a point counts) and temporal support ≤ 5 days; a derived product only where it carries raw observations otherwise inaccessible; monthly only where critical, and named as an exception. One rule is added here because it decides more than the science does: **a source whose terms forbid redistribution is not built**, whatever its value, because every store is published on a public Hugging Face dataset and read by external agents (family 10.1 handover). The registry's `licence.redistribution` field carries **yes**, **attribution** or **no** per group, and a **no** never reaches the Hub.

Two domains. *Terrestrial*: the continents, including inland water and ice sheets. *Coastal*: a band of ± 100 km around the coastline, both sides — the shelf seas where the boundary currents run and the coastal strip where the tide gauges, radars and buoys are. The coastline is the GSHHG shoreline (the Global Self-consistent Hierarchical High-resolution Geography set, public domain, full resolution), rasterised once at

0.05° into a static `dist_coast` channel; “coastal” for a gridded group means the tiles whose distance-to-coast minimum is under 100 km, and for a point store means nothing at all — a point store simply holds every row it has, and the consumer bounds by distance. The band is a parameter of the build, recorded in the registry, not a property of the data.

What is held already and not repeated. Family 7.2’s land channels at 1° (NCEP’s 2 m air, wind, pressure, precipitation, snow, soil moisture and temperature, heat fluxes, skin temperature) and its statics; family 10.1’s tropical moorings and along-track sea level, which run into the coastal band; family 10.2’s fishing effort (E-081, daily 0.1° per vessel, 2012–2024), which is densest on the shelves. Nothing at ≤ 10 km over land exists in any family yet.

2 What the family is for

Honesty first: the four goals — El Niño, the ocean carbon sink, the AMOC, sea-surface temperature — are ocean goals, and most of this family is not an input to their physics. It serves them in three ways, and the falsifier in §7 tests each.

role	mechanism	stores
observed forcing over the land that borders the oceans	the atmosphere the reanalysis was fitted to: hourly pressure, wind and temperature at stations, the tropical upper air from radiosondes, snow and soil state that set continental heating; a model that has only seen the reanalysis cannot learn where it fails (the ICOADS argument of family 1.gf, on land)	<code>ghcnh</code> , <code>ghcnd</code> , <code>igra</code> , <code>lst05</code> , <code>snow05</code> , <code>ims1k</code> , <code>chirps05</code>
coastal sea level and currents as AMOC and El Niño read-outs	the sea-level difference along the US east coast is a Florida Current and Gulf Stream signal at hourly cadence since 1846; Pacific island gauges see Kelvin waves weeks before Niño-3.4 moves; high-frequency radar measures the surface current at 6 km hourly where the boundary currents meet the shelf	<code>tide</code> , <code>hfr</code> , <code>ndbc</code> , <code>gliders</code> , <code>alticap</code>
the read-out side of “predict everything”	Chris (2026-08-19): a predictor of the whole state, from which anything is read out. Fires, land carbon exchange, river flow, snow and vegetation are what a whole-Earth embedding should predict at fine scale — targets first, inputs second, the same argument E-076 made for the ocean biosphere	<code>fire</code> , <code>flux</code> , <code>river</code> , <code>refl05</code> , the tile catalogue and its tokens

Table 1: Three roles, and the stores that fill them.

3 The ledger

Checked 2026-09-16 against producers’ pages, the NASA Common Metadata Repository, and live S3 and directory listings; row counts marked *measured* were summed from the producer’s own inventory files. A dagger (†) marks an estimate that could not be verified.

store	source (producer)	points / grid	cadence	record	rows or pixels	access	licence	republish?
<i>Point networks (tier P)</i>								
<code>ghcnh</code>	GHCN-hourly v1, the successor of ISD (frozen 2025-08-29): temperature, dew point, pressure, wind, precipitation, snow depth, cloud, visibility (NOAA NCEI)	38,870 stations, 10,313 active (measured)	hourly / synoptic	1700s → now, daily updates	7.60×10^9 reports (measured)	S3 <code>noaa-ghcnh-pds</code> , HTTPS; none	CC0	yes

store	source (producer)	points / grid	cadence	record	rows or pixels	access	licence	republish?
ghcnd	GHCN-Daily v3.34: TMAX, TMIN, PRCP, SNOW, SNWD (+139 other elements) (NOAA NCEI)	132,437 stations, 42,984 active (measured)	daily	1763 → now	3.2×10^9 element rows, 112 GB CSV (measured)	S3 <code>noaa-ghcn-pds</code> ; none	CC0	yes
igra	IGRA v2.2 radiosondes: pressure, height, T, RH, dew-point depression, wind at mandatory and significant levels (NOAA NCEI)	2,932 stations, 807 active (measured)	00 / 12 UTC	1905 → now	52.3 M soundings, $\approx 2.1 \times 10^9$ levels; 27.9 GB zipped	HTTPS per station; none	US government	yes
tide	GESLA-4 (its 4.1 download of July 2026) hourly-or-finer sea level, 36 sources incl. UHSLC Fast Delivery (325 gauges, to 2026-07), UHSLC Research Quality (563, to 2024-12), NOAA CO-OPS	6,474 records, 127,623 station-years	$\leq$ hourly	1800s → 2026	$1.1\text{--}2 \times 10^9$ †; 69 GB unzipped	UHSLC ERDDAP (<code>global_hourly_gesla</code>), zip; none	mixed per source: CMEMS, CV, UZ research-only; South African gauges need SANHO permission	per-source filter
ndbc	NDBC standard meteorological historical files, buoys and C-MAN: wind, gust, wave height/period/direction, pressure, air T, SST, dew point (NOAA)	1,345 stations, 829 with 2025 data; 17,043 station-years (measured)	hourly, 10-min recently	1970 → 2025	$3\text{--}5 \times 10^8$ †; 6.2 GB gz	HTTPS station-year; none	US government	yes (TAO sites excluded: held in <code>gtmba</code>)
hfr	HFRNet US national network hourly total vectors at 6 / 2 / 1 km (West, East+Gulf, Puerto Rico, Alaska, Hawaii) + European HFR Node (22 systems)	≈ 12.5 k valid 6 km cells per hour US; 27 EU datasets	hourly	2012 → (US); ≈ 2009 → (EU)	$3\text{--}4 \times 10^9$ US† + $0.3\text{--}1 \times 10^9$ EU†	UCSD THREDDS (unreachable today; CoastWatch ERDDAP keeps 3 months); EU ERDDAP; none	free / CC BY 4.0	yes, attribution
gliders	IOOS Glider DAC (2,540 datasets, 868 delayed-mode) + OceanGliders GDAC (1,120 deployments): T, S, O ₂ , chlorophyll profiles	$\approx 5 \times 10^6$ profiles†	minutes; profiles every ≈ 3 h	2003 →	$2\text{--}5 \times 10^9$ samples†	ERDDAP, SEANOE snapshots (23–25 GB); none	“without restriction” / CC BY 4.0	yes
flux	FLUXNET Shuttle (operational 2026-04-30): 700+ eddy-covariance sites from AmeriFlux, ICOS, TERN, ChinaFLUX; NEE, GPP, latent and sensible heat, radiation, met	700+ sites, $\approx 6,000$ site-years	half-hourly	1991 → 2026	$\approx 1.05 \times 10^8$	<code>data.fluxnet.org</code> ; portal	CC BY 4.0 (six FLUXNET2015 Tier-2 sites excluded; AmeriFlux Legacy sites excluded)	yes
fire	FIRMS active fire: MODIS C6.1 (1 km, 2000-11 →), VIIRS S-NPP (375 m, 2012 →), NOAA-20 (2020 →), NOAA-21 (2024 →): brightness, FRP, confidence, day/night (NASA)	detections	per overpass	2000-11 →	MODIS 1.1×10^8 , VIIRS S-NPP 2.6×10^8 (measured 2023: 4.66 M / 21.75 M), N20+N21 $\approx 1.5 \times 10^8$ †	yearly country zips, no login (to 2024, land only); full archive by request	NASA open†	yes, attribution
river	USGS NWIS daily mean discharge (parameter 00060) + GRDC-Caravan open subset (5,356 catchments, 25 countries)	24,976 US sites, 8,691 active (measured); 5,356	daily	1800s →	$\leq 3.0 \times 10^8$ + $\approx 10^8$	USGS OGC API (key for bulk); Caravan zip	public domain / CC BY 4.0	yes

store	source (producer)	points / grid	cadence	record	rows or pixels	access	licence	republish?
alticap	ALTICAP coastal altimetry, Jason-3 at 20 Hz (≈ 350 m along track), 0–500 km from coast, 66°S–66°N (AVISO/ODATIS)	along-track	≈ 10 -day repeat	2016-02 $\rightarrow$ 2021-07 only; Jason-2 and Sentinel-6 extensions planned	small	AVISO	CC BY 4.0	yes
<i>Daily fields (tier G-fine)</i>								
1st05	MOD11C1 / MYD11C1 v061 land-surface temperature day and night with QC, view time and angle, emissivity; 0.05° 7200 $\times$ 3600 (LP DAAC); continuity VJ121C1 / VJ221C1 (NOAA-20 2018 $\rightarrow$, NOAA-21 2023 $\rightarrow$)	25.9 M px	daily, 2 passes	Terra 2000-02-27 $\rightarrow$, Aqua 2002-07-04 $\rightarrow$ (both still producing; end late 2026 / 2027)	≈ 45 MB/day; 9,591 Terra granules	Earthdata Cloud, Earthdata login, S3 us-west-2	NASA open	yes
ref105	MOD09CMG / MYD09CMG v061 daily surface reflectance bands 1–7, brightness temperatures 20/21/31/32, state and cloud QA; 0.05° (LP DAAC); continuity VJ109CMG / VJ209CMG	25.9 M px	daily	2000-02-24 $\rightarrow$	≈ 600 MB/day, ≈ 5.5 TB per satellite native	Earthdata login	NASA open	yes
snow05	MOD10C1 / MYD10C1 v61 daily snow-cover fraction, cloud fraction, clear index, QA; 0.05° (NSIDC); continuity VNP10C1 / VJ110C1	25.9 M px	daily	2000-02-24 $\rightarrow$	4–6 MB/day	NSIDC, Earthdata login	cite	yes
ims1k	NOAA IMS daily Northern-Hemisphere snow and ice chart, 1 km (24024 ² polar stereographic; 4 km from 2004, 24 km from 1997) (US National Ice Center / NSIDC G02156)	577 M px (1 km)	daily	1 km 2014-12-02 $\rightarrow$	2.8 MB/day gz	HTTPS, no login	cite	yes
chirps05 (E4)	CHIRPS v3.0 pentad precipitation, 0.05°, land 60°S–60°N, infrared + climatology + gauges (UCSB Climate Hazards Center)	17.3 M px	pentad (month-based: six per month)	1981 $\rightarrow$	$\approx 3,300$ pentads	CHC HTTPS, no login	CC BY 4.0	yes
<i>Imagery (tier T: catalogue now, tokens later)</i>								
cat_s2	Sentinel-2 L2A, 10 / 20 / 60 m, 110 km tiles; S2B + S2C (S2A retired 2026-03) (ESA / CDSE)	12.7–15.5 k scenes per day (measured Feb–Aug 2026), ≈ 600 MB each	5-day	2015-07 $\rightarrow$ (global L2A 2018-12 $\rightarrow$)	≈ 5 M scenes/yr, ≈ 3 PB/yr	CDSE STAC / S3 (account; 12 TB per 30 days quota)	Copernicus open	catalogue yes; pixels never stored
cat_s1	Sentinel-1 IW GRDH, 10 m pixels; S1C + S1D (S1A retired 2026-06-30)	982 scenes/day (measured), 1.72 GB each	6-day	2014-10 $\rightarrow$	≈ 360 k/yr, ≈ 620 TB/yr	CDSE	Copernicus open	catalogue
cat_hls	HLS v2.0 S30 + L30, 30 m harmonised Sentinel-2 and Landsat 8/9 (LP DAAC)	S30 2.71 M + L30 1.75 M granules in 2025 (measured)	2–3 day	2013-04 / 2015-11 $\rightarrow$	≈ 0.7 PB/yr; 37.9 M granules so far	Earthdata login, S3 us-west-2	NASA open	catalogue
cat_landsat	Landsat 8/9 Collection 2 L2, 30 m reflectance and surface temperature (USGS)	$\approx 1,400$ –1,500 scenes/day [†]	8-day combined	2013-04 / 2021-10 $\rightarrow$	0.3–0.5 PB/yr [†]	AWS usgs-landsat requester-pays; M2M (login)	public domain	catalogue

store	source (producer)	points / grid	cadence	record	rows or pixels	access	licence	republish?
cat_nisar	NISAR L2 GCOV provisional (2026-07-20 →; beta 2025-10 → 2026-01), L-band, 10 / 20 / 80 m posting (ASF)	28,456 granules in August 2026, 0.26–7.7 GB each	12-day (gap 2026-07-27 → 08-10)	2026-06-17 →	0.7–1 PB/yr	ASF, Earthdata login	NASA open [†]	catalogue
cat_olci	Sentinel-3 OLCI L2 WFR, 300 m water-leaving reflectance and chlorophyll, coastal band only; and GlobColour OLCI 300 m L3 (Copernicus 009.103) as the gridded form	457 scenes/day, 150 GB/day (measured)	1–2 day	2016-04 →	≈ 55 TB/yr	CDSE / Copernicus Marine	Copernicus open	catalogue; coastal tokens
cat_viirs	VIIRS I-band L1B 375 m (VNP02IMG / VJ102IMG), 6-minute granules (LAADS)	225–242 granules/day, ≈ 170 MB	daily	2012 / 2018 →	≈ 15 TB/yr per satellite	LAADS S3, Earthdata login	NASA open	catalogue
cat_dswx	OPERA DSWx-HLS (2016 →) and DSWx-S1 (2023-12 →), 30 m surface-water class; DIST-ALERT-HLS (2023 →)	3.1 M / 2.5 M / 4.0 M granules per year (measured)	2–12 day		90 / 14 TB/yr	PO.DAAC / LP DAAC	NASA open	catalogue

Table 2: The candidates that pass the rule and the licence gate, as checked on 2026-09-16.

Checked and refused by the licence gate. **ISMN** (3,300 soil-moisture stations, hourly, 1952 →): “re-export or transfer of the original data to a third party is prohibited”. **GRDC** (10,000 river gauges): “redistribution to the general public (incl. distribution via Internet) is not allowed” — only its CC BY 4.0 Caravan subset is taken. **ECA&D** (25,919 European stations): non-commercial, no redistribution; **GHCN-Daily** carries most of the same stations under CC0. **HadISD** (10,405 stations, final version 2025-08): Non-Commercial Government Licence. **MSWEP** precipitation: CC BY-NC. None is built. **ECOSTRESS** (70 m thermal from the ISS, irregular) is open and passes the resolution rule but has no cadence; it is catalogued with the tiles rather than stored.

Checked and set aside on the rule. **SMAP** 9 km enhanced soil moisture (SPL3SMP.E) is a Backus–Gilbert resampling of a ≈ 40 km footprint and the 36 km product is the observation; neither passes, and the **ASCAT** 12.5 km / 25 km-effective record (H SAF, CC BY 4.0) does not either — soil moisture from space has no ≤ 10 km observation, and the in-situ network cannot be published, so the family has *no soil-moisture store*; family 7.2’s 1° NCEP channel stays the only one. **Copernicus Land 300 m LAI/FAPAR** is 10-daily and derived, and 32 TB as 5-day dense frames; refused. **MCD43C4** (0.05° daily NBAR reflectance) is a 16-day BRDF inversion; the raw daily MOD09CMG is taken instead. **Hydroweb** river and lake heights (11,336 virtual stations, ETALAB open) revisit every 10–35 days; refused on cadence. **IMERG** at 0.1° is family 1.gf’s exception E2 and one global store, not repeated here. **ESA CCI soil moisture** at 0.25° and **GRACE** at 300 km are far outside the rule and not critical to the four goals. **ACSPO L3S 0.02° SST** and **MUR** 1 km are global and belong to 1.gf (sst_acspo02); MUR is a gap-filled analysis and is not taken anywhere. **SNOTEL** (932 snow stations) is largely inside **GHCN-Daily**’s SNWD and WESD elements and is not a separate store.

4 Storage

Everything of family 1.gf §4 applies: the registry at `tensors/family1_tf/family1tf.json` (schema 2, the tier letters G, P, T with `layout` distinguishing bin-major, sharded and static grids, the new per-group fields `licence`, `frames_per_bin`, `frame_seconds`, `dtype` and `verdict`, and the `siblings` line in family 10.2’s registry, the current family-10 registry that inherits 10.1’s stores and adds E-081’s `fishing`); tier P from family 10.1 with 1.gf’s one widening (**schema 3**, int64 `time_s`, 31 + 2C bytes a row, for every store whose record starts before 1914 — here `ghcnh`, `ghcnd`, `igra`, `tide`, `river`); the sharded tiled `zstd` layout for grids finer than 0.25°; two-machine builds with credentials only on hosted runners; and no silent-skip path. Three things are specific to land and coast.

Station stores are wide rows keyed by platform, and the platform table is part of the store.

A station’s row is one observation time with C channels (`ghcnh`: temperature, dew point, sea-level pressure, wind u and v , precipitation, snow depth, cloud fraction — $C = 8$; `ghcnd`: TMAX, TMIN, PRCP, SNOW, SNWD — $C = 5$), `platform` the station’s numeric hash, and the store ships one file beyond family 10.1’s ten, `platforms.json`: for each hash the source identifier, elevation, and the source’s own station metadata, because a station’s elevation is what makes its temperature comparable to a 0.05° skin temperature and it must not be lost in the hash. `qc` carries the source’s own flag; nothing is homogenised. GHCN-Daily’s long format (one row per station, day, element) is pivoted to one row per station-day; `ghcnh`’s synoptic reports keep their own minute in `time.s`.

Profiles and soundings put the vertical axis in C . `igra` holds one row per sounding with $C = 80$: temperature, dew-point depression, u , v and geopotential height at the 16 mandatory pressure levels (1000 to 10 hPa), NaN above the balloon’s burst; significant levels are not kept in phase A. `gliders` holds one row per profile with the 16 Roemmich–Gilson pressures for T, S, O_2 and chlorophyll ($C = 64$), the same convention as family 8 and family 1.gf’s `wod`.

The tile catalogue is a tier-P store with nothing in values but quality. One row per scene-date: `bin`, `time.s` (acquisition), `lat`, `lon` (footprint centre), `platform` the hashed scene identifier, `qc` the producer’s processing-baseline flag, `fp` the footprint (a 110 km Sentinel-2 tile is $\log_2(110/27.83) = +2.0$; a 30 m pixel is -9.9 — the catalogue carries the *tile*’s, the token cache will carry the token’s), and $C = 4$: cloud fraction, valid-pixel fraction, sun zenith (or radar incidence angle), and the footprint’s area in km^2 . A sidecar `assets.parquet` maps the hash to the STAC identifier and the asset URLs, so the token builder of E-078c can fetch exactly the scenes a coastal cell needs. The catalogue for all eight sensors is ≈ 150 M rows over 2013–2026, 6 GB — cheap enough to build in phase A and the only way to plan the token build against measured scene counts rather than the producers’ round numbers. The token cache itself (E-078 §4: $k = 4$ –8 tokens of 64 dimensions per tile-date, from a local codec conditioned on sensor metadata) is a later tier-P store, first over the ≈ 300 coastal 0.25° cells of E-078c; its size is measured, not designed, by that experiment.

Grids that are not lat/lon. `ims1k` is on a polar stereographic 24024×24024 grid; like family 1.gf’s sea ice it is stored on its native grid with the projection in `tile.grid.json` and the tile corners in latitude and longitude, so the reader’s cone search needs no reprojection — a tile is found by its corners, a pixel’s position by the projection formula the registry names.

5 The stores in numbers

Estimates in the layout above; the probe of §7 replaces every one with a measurement. P bytes are $(27 + 2C)$ per row; G-fine bytes are $\text{pixels} \times \text{frames} \times v \times 2C \times 1.2$, with v the valid fraction (land fraction times clear-sky fraction for the optical products).

Two arithmetic notes. `ghcnh` at 7.6 billion reports is the largest point store the programme would have built, three and a half times `slatrack`’s rows; it is phase B so that `ghcnd`, an eighth of the size, can show first whether station observations move any goal at all. `hfr`’s total is unverified because the UCSD THREDDS archive refused connections on the day of checking, and the CoastWatch mirror keeps only three months; the probe must reach the archive before the store is sized.

6 The exceptions, named

- **E4 — CHIRPS pentads.** Derived (infrared cold-cloud duration plus a climatology plus gauges), not an observation; admitted because the gauge record it carries for Africa and South America is not otherwise reachable under an open licence (GHCN-Daily is thin there), and because it is 0.05° and pentad-native, so it fits the family’s time axis with its own six-per-month bin table in the registry. CC BY 4.0.
- **Monthly: none.** Nothing monthly is critical here; PSMSL’s monthly tide-gauge series (1,620 stations, 1807 $\rightarrow$) is covered by GESLA’s hourly record for every station that matters to the goals.

store	tier	channels (C)	footprint ($\log_2 f_p, \log_2 f_t$)	rows or frames	stored	role	phase
ghcnd	P	5: TMAX, TMIN, PRCP, SNOW, SNWD (schema 3)	$(-4, -2.32)$	$\approx 1.0 \times 10^9$ station-days (3.2 B element rows)	≈ 40 GB	forcing	A
igra	P	80: T, Td dep., u, v, z at 16 mandatory levels	$(-4, -4)$	52.3 M soundings	10 GB	forcing, El Niño	A
tide	P	1: sea level (mm, relative) (schema 3)	$(-4, -6.9)$	$1.2\text{--}1.5 \times 10^9$ after the licence filter	40–50 GB	AMOC, El Niño	A
ndbc	P	10: wind u, v , gust, Hs, Tp, Tm, dir, pressure, air T, SST	$(-4, -6.9)$	$3\text{--}5 \times 10^8$	15–25 GB	SST, forcing	A
gliders	P	64: T, S, O ₂ , chl at 16 pressures	$(-4, -4)$	$\approx 5 \times 10^6$ profiles	0.8 GB	AMOC, SST, carbon	A
flux	P	10: NEE, GPP, Reco, LE, H, Rn, Ta, VPD, SWC, P	$(-4, -7.9)$	$\approx 1.05 \times 10^8$ half-hours	5 GB	read-out (land carbon)	A
fire	P	4: brightness, FRP, confidence, day/night	$(-4.8 / -6.2, -4)$	$\approx 5 \times 10^8$	20 GB	read-out, carbon	A
river	P	1: discharge (schema 3)	$(-4, -2.32)$	$\approx 4 \times 10^8$	14 GB	read-out, AMOC (freshwater)	B
alticap	P	2: sea-level anomaly, quality	$(-6.3, -4)$	$\approx 10^{8\dagger}$	3 GB	AMOC	B
ghcnh	P	8: T, Td, SLP, u, v , precip, snow depth, cloud (schema 3)	$(-4, -6.9)$	7.6×10^9	360 GB	forcing	B
hfr	P	2: u, v at the cell centre; platform = network	$(-2.21 / -3.8, -6.9)$	$3.5\text{--}5 \times 10^9$	110–155 GB	AMOC (Florida Current edge), read-out	B
1st05	G-fine	4: LST day, LST night, QC day, QC night ($F = 5$; both passes as channels of the day's frame; view time as 2 more in phase B)	$(-2.32, -2.32)$	25.9 M px $\times$ 9,700 d; $v \approx 0.29 \times 0.5$	≈ 350 GB per satellite; Terra first	forcing, SST's land twin	A
snow05	G-fine	2: snow fraction, cloud fraction, stored as uint8 percent	$(-2.32, -2.32)$	25.9 M px $\times$ 9,700 d; $v \approx 0.29$; classes compress far below the float formula (the producer's files are 4–6 MB/day)	≈ 60 GB (Terra)	forcing	A
cat_*	P	4: cloud, valid, sun/incidence, area	tile footprint	$\approx 1.5 \times 10^8$ scene-dates	6 GB	the T tier's plan geometry	A
static_fine	G-static	GEBCO_2026 15'' elevation and bathymetry (7.5 GB float16); WorldCover 2021 class fractions at 0.05° (11 ch, 0.6 GB); dist.coast at 0.05°	—	one frame	≈ 8 GB		A
ref105	G-fine	8: bands 1–7, state QA	$(-2.32, -2.32)$	25.9 M px $\times$ 9,700 d; $v \approx 0.15$	≈ 700 GB (Terra); VJ1 2018 $\rightarrow$ adds ≈ 250 GB	read-out (vegetation, snow, burn), albedo	B
ims1k	G-fine	1: class (uint8: water, land, ice, snow)	$(-4.8, -2.32)$	577 M px $\times$ 4,300 d; classes compress $\approx 50 \times$	≈ 50 GB	forcing	B
chirps05 (E4)	G-fine	1: precipitation (mm/pentad); CHIRPS's own six-per-month pentads carried as a frame_table of (start, end) seconds, each frame filed under the family bin holding its midpoint	$(-2.32, 0)$	17.3 M px $\times$ 3,300; $v \approx 0.3$ (land in the box)	≈ 40 GB	forcing, El Niño teleconnection read-out	B
1st05 Aqua, VIIRS	G-fine	as above			$\approx 350 + 160$ GB	continuity past Terra	B
tile tokens (E-078c)	P	$k \times 64$ tokens per tile-date	tile	coastal cells first	measured by E-078c	the T tier	C
oc300_coast	G-fine	2: chl, flags at 300 m in the 100 km coastal band	$(-6.5, -2.32)$	$\approx 6 \times 10^8$ px $\times$ 3,800 d; $v \approx 0.15$	≈ 1.6 TB	carbon (coastal), read-out	C
sst02_coast	G-fine	the coastal band of 1.gf's sst_acspo02, if that store is not built whole			≈ 0.3 TB	SST	C

Table 3: Estimated size of every store, and its phase. Phase A totals ≈ 0.6 TB (of which 1st05 Terra is 350 GB and the seven point stores together ≈ 140 GB); phase B ≈ 2 TB more; phase C waits on E-078c.

7 The build ladder, the probe, and the falsifier

Phase A — eleven stores, ≈ 0.6 TB. `ghcnd`, `igra`, `tide`, `ndbc`, `gliders`, `flux`, `fire`, `1st05` (Terra), `snow05` (Terra), the eight `cat_*` catalogues (one store per sensor, one adapter each, counted as one here), `static.fine`, and the registry. Seven of the eleven fetch without any account (`ghcnd`, `igra`, `tide`, `ndbc`, `gliders`, `flux`, `static.fine`); `1st05`, `snow05`, `fire` (the NOAA-20/21 archive) and the NASA catalogues need an Earthdata login on a hosted runner; the Sentinel catalogues need a CDSE account for STAC, and the Landsat one a USGS login or a requester-pays AWS read. Cost from the 10.1 precedent: the point stores \$0 on hosted runners; `1st05` one verified box for a few hours to assemble 350 GB of shards, \$2–5; under \$10 for the phase.

Phase B — ≈ 2 TB. `ghcnh`, `hfr`, `river`, `alticap`, `refl05`, `ims1k`, `chirps05`, and the Aqua and VIIRS continuations of `1st05` and `snow05`. \$20–30 in box time; the Hub conversation about terabyte repositories before it, as for family 1.gf.

Phase C — after E-078c. Tile tokens over the coastal cells; `oc300_coast`; the coastal SST band if 1.gf’s global store is not built.

The probe. One month per source through the real adapter on a hosted runner: rows per day and distinct platforms (point stores); valid fraction and compressed bytes per tile (G-fine); scene counts per day against the producer’s catalogue (the `cat_*` stores); throughput. For `tide`, the probe also runs the per-source licence filter and reports how many station-years it removes, because that number, not the archive’s, is what the store will hold.

The falsifier. Two, because the family has two kinds of store. For the *forcing and coastal* stores, family 1.gf’s test applies unchanged: at anchors where the store has coverage, its tokens in the cone must lower the held-out masked-token loss on the goal’s read-out against the same anchors without them, three seeds; parity means the reanalysis and the coarse tensor already carried it. For the *read-out* stores (`fire`, `flux`, `river`, `refl05`, the tokens), the test is the other way round: the store is a *target* of E-078 §5’s footprint-queried decoder, and it earns its place if the embedding trained without it predicts it better than climatology at its own footprint — the E-076b form of the question (optimal interpolation from the same observations is the bar). A read-out store that cannot be predicted from the embedding says the embedding is not yet generic; it is not evidence against the store.

8 Decisions for Chris

1. **The name** — as given; if “1” means “10”, it is a prefix change. Pick: as given.
2. **The licence gate:** never publish ISMN, GRDC (except Caravan), ECA&D, HadISD, MSWEP; filter GESLA per source. Pick: yes — and the consequence, that the family has no soil-moisture store, is accepted rather than patched with a 36 km product.
3. **The coastal band:** ± 100 km. Pick: 100 km; the shelf break is inside it almost everywhere and the radars reach 200 km at most.
4. **Daily fields at 0.05° :** Terra first, Aqua and NOAA-20/21 in phase B; MODIS Collection 7 replaces v061 when it is published. Pick: yes.
5. **ghcnh in phase B**, after `ghcnd`’s verdict. Pick: yes.
6. **E4 (CHIRPS):** admit. Pick: admit; it is the one derived *field* admitted as an input (IMS is an analyst’s chart and the DSWx and WorldCover classifications are read-outs or statics), and it carries gauges.
7. **The catalogue before the codec:** build all eight `cat_*` stores in phase A even though the tokens wait on E-078c. Pick: yes — 6 GB that turns the tile tier’s scene counts into measured numbers.
8. **Tide gauges:** GESLA-4.1 through the UHSLC ERDDAP, filtered, or UHSLC’s own two sets only (325 + 563 gauges, licence clean). Pick: GESLA filtered; the pick reverses to UHSLC-only if the filter cannot be applied per station.

Terms used above, in one line each

Tier G / P / T family 10's three storage tiers: gridded fields; points, profiles and tracks; tiles of imagery held as a catalogue plus codec tokens. **STAC** SpatioTemporal Asset Catalog, the standard in which producers list scenes. **zstd** Zstandard, a fast lossless compressor. **GSHHG** the Global Self-consistent Hierarchical High-resolution Geography shoreline set. **GHCNh / GHCN-Daily / ISD** NOAA's hourly and daily station archives and the frozen hourly archive GHCNh replaces. **IGRA** the Integrated Global Radiosonde Archive (weather balloons). **GESLA / UHSLC / CO-OPS / PSMSL** the global hourly tide-gauge compilation, the University of Hawaii Sea Level Center, NOAA's tide-gauge service, and the monthly Permanent Service for Mean Sea Level; **CV, UZ, CMEMS** three GESLA source codes with research-only terms; **SANHO** the South African Navy Hydrographic Office. **NDBC / C-MAN** NOAA's buoy centre and its coastal automated stations. **HFRNet / IOOS** the US high-frequency coastal radar network and the Integrated Ocean Observing System that runs it and the glider data-assembly centre (DAC). **SEANOE** Ifremer's data-publication service. **FLUXNET / ICOS / AmeriFlux** networks of eddy-covariance towers measuring the exchange of CO₂, water and heat between land and air; **NEE / GPP / Reco / LE / H / Rn / VPD / SWC** net ecosystem exchange, gross primary production, ecosystem respiration, latent and sensible heat, net radiation, vapour-pressure deficit, soil water content. **FIRMS / FRP** NASA's fire-detection service and fire radiative power. **NWIS / OGC** the US Geological Survey's water archive and the Open Geospatial Consortium API it now uses. **ALTICAP / ODATIS / AVISO** a coastal altimetry product and the French ocean-data services that host it. **CMG MODIS' Climate Modeling Grid, 0.05°**. **IMS** the US National Ice Center's daily snow and ice chart. **BRDF / NBAR** the bidirectional reflectance model and the nadir-adjusted reflectance derived from it (a 16-day inversion, so derived). **L1B / L2 / L2A / L3** calibrated radiances; retrievals on the swath; Sentinel-2's atmospherically corrected surface reflectance; gridded. **CDSE** the Copernicus Data Space Ecosystem, ESA's Sentinel archive. **IW GRDH** Sentinel-1's interferometric-wide-swath ground-range-detected product. **HLS** Harmonized Landsat Sentinel-2, NASA's 30 m merge of the two. **GCOV / ASF** NISAR's geocoded covariance backscatter product and the Alaska Satellite Facility that serves it. **OLCI WFR** Sentinel-3's ocean-colour instrument and its water full-resolution product. **LAADS** NASA's atmosphere archive that serves VIIRS radiances. **DSWx / DIST-ALERT** NASA OPERA's surface-water and vegetation-disturbance maps. **ISMN / ECA&D / HadISD / MSWEP / GRDC** the soil-moisture network, the European station set, the Met Office hourly station set, a merged precipitation product, and the Global Runoff Data Centre — all refused on licence. **H SAF** EUMETSAT's hydrology facility. **LAI / FAPAR** leaf-area index and the fraction of absorbed photosynthetic radiation. **ETALAB** the French open-data licence. **SNWD / WESD** GHCN-Daily's snow-depth and snow-water-equivalent elements. **GEBCO** the General Bathymetric Chart of the Oceans. **M2M** the USGS machine-to-machine API. **Td / SLP / Hs / Tp / Tm** dew point, sea-level pressure, significant wave height, peak and mean wave period. **Caravan** the CC BY subset of GRDC gauges published with the Caravan hydrology dataset. **E-076 / E-076b / E-078c** family 8's nearest-observation design, its optimal-interpolation measurement, and the first coastal tile-codec experiment of E-078.

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